Manifold for T1D – Joslin Preservation & Medalist Analytics

β-cell preservation, Medalist residual-function, and honeymoon-phase response prediction

The Joslin Diabetes Center holds two assets no other institution matches at comparable depth: the **Medalist cohort** — the world's most direct empirical demonstration that β-cell function can persist under continuous autoimmune pressure — and one of the largest **new-onset and honeymoon clinical populations** in the United States, now with approved preservation agents (teplizumab, verapamil, low-dose ATG, GLP-1 adjuncts). **Manifold** configures three engine capabilities to the Joslin data: causal discovery on multi-omic plus longitudinal clinical data, criticality estimators (nonlinear mixed-effects C-peptide, HTE learners, BOCPD, long-horizon survival) feeding a composite fragility score (**Ω**F), and Pearl do-calculus with interdiction for principled preservation-therapy selection.

> The Medalist observation — persistent measurable C-peptide fifty years into T1D — is a stage-change in how the field conceives of β-cell reserve. The outstanding question is no longer whether residual function persists, but what distinguishes the patients in whom it does — a problem of causal inference.

ΩF PILLAR MAPPING – JOSLIN PRESERVATION PHYSIOLOGY

ΩF PILLAR	JOSLIN READING
I — Interaction	Autoantibody × HLA × drug-exposure interactions for preservation response; Medalist-phenotype genotype × metabolic-history interactions
R — Reserve	C-peptide AUC on MMTT as direct β-cell functional-mass proxy; Medalist long-horizon residual-function signature
J — Junction	Immune–metabolic coupling: autoantibody titer ↔ C-peptide transfer entropy; cytokine-panel ↔ glycemic-variability coupling
C — Cascade	Autoreactive T-cell clonal expansion; epitope-spreading rate; β-cell senescence in Medalist tissue
T — Temporal	Hazard of honeymoon-end given current trajectory; time-to-loss-of-residual-function in Medalist subjects

PROPOSED PARTNERSHIP

Partnership with a Joslin Medalist-study or Clinic-affiliated principal investigator is sought for:

1. Retrospective Clinic and Medalist data access under Joslin's existing Data Use Agreement and IRB.
2. Clinical co-authorship on the preservation-HTE / Medalist-phenotype-classification benchmark manuscript (twelve to fourteen weeks from data access).
3. Co-application to Breakthrough T1D, the Helmsley Charitable Trust, or NIH NIDDK for the Phase 2 prospective arm.

Apex Analytica contributes the causal-inference engine, the **Ω**F framework, and the engineering team. Full brief: `M-T1D-02-JOSLIN`.